

Literature & setup:

Literature Review:

Alain & Bengio (2016)

This foundational paper provides the methodological justification for using linear probes to measure how "explicitly" a concept is represented within a model's hidden layers.

Definition of a Probe: A linear classifier (probe) is added to the hidden units of an intermediate layer to measure how much information that layer contains about a specific target.

Information vs. Computational Convenience: While deep layers technically destroy Shannon information, they perform transformations that make information immediately useful to a linear classifier.

Application to My Project: High probe accuracy at a specific layer suggests that the model has "massaged" the data into a representation where physical concepts (like structural stability) are linearly separable and thus explicitly encoded

Hummel et al. (2026)

This study demonstrates how linear probing can isolate specific, domain-relevant features from complex, "noisy" pretrained embeddings.

Feature Isolation: The paper shows that even if an embedding space is dominated by irrelevant data (such as recording-specific IDs), a simple linear probe can effectively suppress this noise to isolate relevant features (such as ship types).

Comparison of Models: It systematically compares general-purpose audio, speech, and domain-specific bioacoustic models using linear probing to assess their generalizability to underwater acoustics.

Application to My Project: It provides a template for comparing domain-specific vs. general-purpose models. It supports your goal of determining if materials science LLMs isolate physical concepts more effectively than general models.

Rubungo et al. (2023)

This paper (LLM-Prop) directly compares general-purpose models to domain-specific materials science models for predicting physical properties like band gaps and unit cell volume.

General vs. Specific Performance: Surprisingly, the researchers found that a general-purpose model (T5 encoder) with a simple linear layer could outperform domain-specific models like MatBERT and state-of-the-art Graph Neural Networks (GNNs).

Internal Knowledge of Symmetry: The model's success in predicting unit cell volume suggests it can access critical physical information, such as space group symmetry, more effectively from text descriptions than specialized structural models.

Application to My Project: This serves as a primary reference for my overall question. It suggests that general models may already encode significant "domain-relevant concepts" if they have been exposed to scientific literature during pretraining.

Zhang & Yang

This study on polymer property prediction (PolyLLMem) explores whether LLMs truly "internalize" chemical understanding or simply mimic patterns.

Visualizing Internal Representations: Using UMAP to project Llama 3 embeddings, the authors found that the model naturally forms clusters based on physical properties (e.g., glass transition temperature and band gap) prior to any task-specific training.

Token-Level Interpretability: By analyzing token-level similarity, the researchers demonstrated that Llama 3 internalizes understanding of chemical symmetry and structural relationships (e.g., recognizing aromatic rings and the impact of fluorine substituents).

Application to My Project: This supports my overall motivation. It proves that general-purpose LLMs like Llama 3 can effectively capture both syntactic structure and underlying chemical relationships.

